A Composition-Controlled Evaluation of RNA Structure Awareness in Foundation Models

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Abstract

RNA foundation models are entering drug-discovery pipelines, where their structural confidence scores guide target selection for screening campaigns. Systematic controls separating learned RNA structure from nucleotide composition and architectural inductive bias are largely absent from current evaluations. We present a pre-registered, composition-controlled evaluation of seven RNA and DNA foundation models across 16 therapeutic RNA targets spanning eight druggable positives, five non-druggable negatives, and three graded HTT repeat-length controls. The evaluation applies a two-stage gate: Stage 1 tests whether each model's mutation sensitivity exceeds a nucleotide-stratified permutation null (1,000 permutations, 95th-percentile threshold); Stage 2 tests whether the model's best-head attention pattern correlates with the ViennaRNA-predicted contact map ($\rho_S \geq 0.10$). The primary pre-registered hypothesis—that the gate discriminates druggable from non-druggable targets better than probing accuracy—fails ($p = 1.0$): the composition null rejects 14–16 of 16 targets per model, including most positives. The gate catches a distinct failure mode the primary hypothesis did not target: two targets pass the composition null but lack attention-contact correspondence (the "degeneracy trap"). A five-seed untrained-weight control reveals that Nucleotide Transformer v2 produces identical attention-contact correlation from random weights (delta $= -0.002$, 95% CI $[-0.011, +0.007]$), exposing a purely architectural artifact invisible to any evaluation that omits untrained controls. The composition-null gate functions as an artifact detector rather than a druggability predictor—a role that existing RNA foundation model evaluations do not fill.

Author summary RNA foundation models are increasingly used to prioritize drug targets: a model assigns a confidence score to a candidate RNA, and that score determines whether the target advances to expensive experimental screening. We tested whether these scores reflect genuine structural understanding or simpler confounds. Using a panel of 16 therapeutic RNA targets with known druggability status, we found that most models' sensitivity to mutations can be explained by nucleotide composition alone—the models respond differently to mutations at stem positions than loop positions, but so would any model that has learned which nucleotides tend to appear in stems. A separate control revealed that one model's attention patterns correlate with RNA structure even when its weights are completely random, an architectural artifact rather than learned knowledge. These findings suggest that per-target composition controls and untrained-weight baselines should be standard in RNA foundation model evaluations. All hypotheses were pre-registered before any data were analyzed.

1 Introduction

RNA foundation models trained on millions of non-coding RNA sequences have achieved strong performance on secondary-structure prediction benchmarks and are now entering therapeutic target-selection pipelines [1–3, 6, 7]. A model's structural confidence score on a candidate RNA target increasingly determines whether that target proceeds to experimental screening—a decision that commits hundreds of thousands of dollars in assay development and compound library costs. Whether a high confidence score reflects learned RNA structure or a confound that mimics it has received limited systematic evaluation.

Two confounds dominate. Stem regions in RNA secondary structures are enriched for G and C (mean GC content ∼68.5%) relative to loop regions (∼43.9%) [8]. A model that has learned to respond to local GC content will show high mutation sensitivity at stem positions and low sensitivity at loop positions—the same pattern as genuine structural awareness—without encoding base-pairing geometry. A nucleotide-stratified permutation null, which shuffles stem and loop labels within each nucleotide type, controls for this confound: in a prior evaluation across 52 Rfam families, RNA-FM's apparent structure sensitivity ratio dropped from ∼5.5× to 1.8× after nucleotide control, and no family survived both nucleotide and dinucleotide nulls [8]. A second, independent confound is architectural inductive bias: transformer attention patterns can correlate with RNA contact maps through positional encoding alone, independent of learned weights, a failure mode invisible to evaluations that test only trained models.

Existing evaluations do not control for either confound at the per-target level. OmniGenome [9] and CHANRG [10] evaluate aggregate accuracy across held-out families—answering whether a model generalizes to new sequences, not whether its confidence on a specific therapeutic target is real. A model can achieve 78% aggregate probing accuracy while passing a composition null on only 1 of 16 targets.

We present a pre-registered, per-target evaluation of seven foundation models (five RNA, two DNA) against 16 therapeutic RNA targets with known druggability status, applying a two-stage composition-controlled gate and an untrained-weight architectural control. Four hypotheses were pre-registered and frozen with SHA-256 hashes before evaluation. The primary hypothesis—that the gate discriminates druggable from non-druggable targets—fails. The gate catches composition artifacts and architectural degeneracy that no existing evaluation detects, functioning as a quality filter on computational evidence rather than a druggability predictor.

2 Methods

Figure 1 summarizes the two-stage gate and untrained-weight control.

2.1 Target benchmark

Sixteen RNA targets were selected to span three categories: eight positives (POS-01 through POS-08) with established druggability evidence, five negatives (NEG-01 through NEG-05) with no known druggability, and three graded controls (CTL-01 through CTL-03) consisting of HTT exon 1 CAG-repeat constructs at lengths 17, 36, and 60, providing a monotonicity test for repeat-length sensitivity. Table 1 lists each target with its RNA type, sequence length, and druggability evidence. The benchmark is stored as `druggability_benchmark_v2.tsv` and was frozen before evaluation.

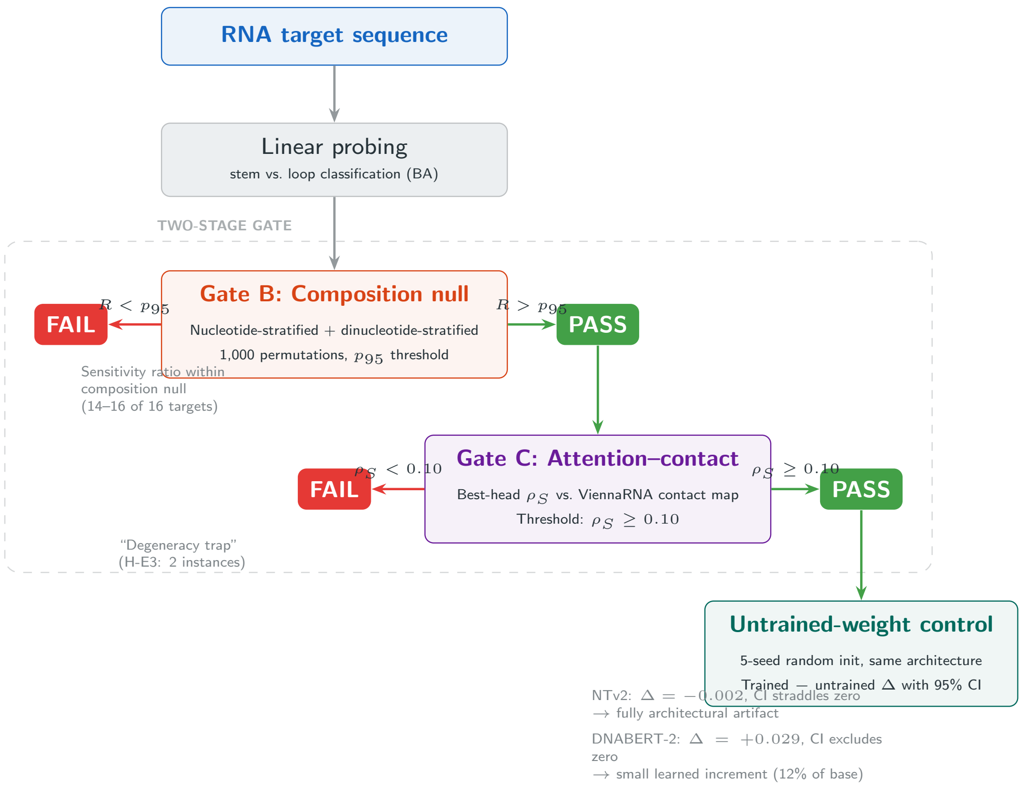

Figure 1: Two-stage composition-controlled evaluation pipeline. Gate B tests whether each model's mutation sensitivity ratio exceeds a nucleotide-stratified and dinucleotide-stratified permutation null (1,000 permutations, 95th-percentile threshold). Targets surviving Gate B proceed to Gate C, which tests attention–contact correspondence ($\rho_S \geq 0.10$). Targets passing Gate B but failing Gate C fall into the "degeneracy trap" (H-E3). The untrained-weight control re-runs Gate C with randomly initialized weights to separate learned representations from architectural inductive bias.

Table 1: Target benchmark. POS: druggable positive; NEG: non-druggable negative; CTL: graded HTT repeat-length control. Druggability evidence lists published ligands or structural rationale. Synthetic controls (NEG-04, NEG-05) are generated at runtime with frozen RNG seeds.

ID	**Target**	**RNA type**	**Source**	**nt**	**Evidence**
POS-01	SMN2 exon 7 splice element	pre-mRNA	NM_017411.3	54	Risdiplam (FDA)
POS-02	FMN riboswitch aptamer	riboswitch	RF00050	78	Ribocil (K_d 16 nM)
POS-03	HCV IRES domain II	viral IRES	PDB 1P5O	77	Cpd 13 (K_d 720 nM)
POS-04	HIV-1 TAR apical loop	viral element	PDB 1ARJ	29	Amiloride derivatives
POS-05	PreQ1 riboswitch aptamer	riboswitch	RF00522	51	Cpd 1 (K_d 99 nM)
POS-06	MALAT1 ENE triple helix	lncRNA	PDB 4PLX	76	Triplex stabilizers
POS-07	EV71 IRES stem-loop II	viral IRES	PDB 6XB7	41	DMA-135 (K_d 500 nM)
POS-08	SARS-CoV-2 5′ UTR SL	viral element	NC_045512.2	47	DMA-155 (IC_{50} 10 μM)
NEG-01	HOTAIR lncRNA domain 1	lncRNA	NR_047517.1	100	R-scape: no covariation
NEG-02	SRA lncRNA 5′ region	lncRNA	NR_045587.1	100	R-scape: no covariation
NEG-03	Xist repeat A tandem	lncRNA	NR_001564.2	111	No conserved structure
NEG-04	Shuffled tRNA-Phe	synthetic	seed 42	76	Dinuc-preserved shuffle
NEG-05	Random GC-matched	synthetic	seed frozen	76	No designed structure
CTL-01	HTT CAG17 (normal)	mRNA repeat	NM_002111.7	235	Repeat-length ref
CTL-02	HTT CAG36 (threshold)	mRNA repeat	NM_002111.7	292	Repeat-length ref
CTL-03	HTT CAG60 (severe)	mRNA repeat	NM_002111.7	364	Repeat-length ref

2.2 Foundation model panel

Seven models were evaluated: RNA-FM [1], RiNALMo [2], ERNIE-RNA [3], SpliceBERT [4], UTR-LM [11], Nucleotide Transformer v2 (NTv2) [12], and DNABERT-2 [5]. Five are RNA models; NTv2 and DNABERT-2 are DNA models included to test cross-domain transfer. All models were used with frozen pretrained weights (no fine-tuning). DNABERT-2's BPE tokenization prevents nucleotide-level mutation mapping, so it is excluded from Gate B (composition-null sensitivity) and included only in Gate C (attention-contact) and probing analyses. For DNABERT-2, BPE token-level hidden states were expanded to nucleotide resolution via proportional mapping (`_expand_bpe_to_nucleotide`), and attention was aggregated to nucleotide-level contacts (`_aggregate_contacts_to_tokens`) using a 6-mer approximation. Table 2 summarizes the model panel.

Table 2: Foundation model panel. Token resolution determines Gate B eligibility: DNABERT-2's BPE tokenization prevents nucleotide-level mutation mapping.

Model	**Architecture**	d	**Layers**	**Token res.**
RNA-FM [1]	Transformer encoder	640	12	nucleotide
RiNALMo [2]	Transformer encoder	1280	33	nucleotide
ERNIE-RNA [3]	Transformer encoder	768	12	nucleotide
SpliceBERT [4]	BERT	512	6	nucleotide
UTR-LM [11]	Transformer encoder	128	6	nucleotide
NTv2 [12]	ESM (attn + GLU)	512	12	6-mer
DNABERT-2 [5]	MosaicBERT	768	12	BPE

2.3 Probing accuracy

For each model–target pair, a linear probe was trained on per-position embeddings (extracted at each model layer) to classify nucleotide positions as stem or loop, using the ViennaRNA-predicted secondary structure as ground truth. Balanced accuracy (BA) = (sensitivity + specificity) / 2 was computed across all positions in all 16 targets, and the best-layer BA was reported per model.

2.4 Gate B: composition-null mutation sensitivity

Each nucleotide position in a target sequence was perturbed by complement swap (A↔U, C↔G, with T treated as equivalent to U for DNA models). The model embedding distance (L2 norm) between wild-type and perturbed sequence was measured at the mutated position, extracted at each model layer. The structure sensitivity ratio R was computed as the ratio of mean embedding distance at stem positions to mean distance at loop positions, maximized across layers:

$$R = \max_{\ell} \frac{\bar{d}^{(\ell)}_{\text{stem}}}{\bar{d}^{(\ell)}_{\text{loop}}} \tag{1}$$

Two null distributions were constructed per target. The *nucleotide-stratified null* shuffled stem/loop labels independently within each nucleotide type (A, U, C, G), preserving the compositional correlation between nucleotide identity and structural role. The *dinucleotide-stratified* *null* further conditioned on the identity of each position's nearest neighbor, controlling for second-order sequence context. Each null was sampled 1,000 times, with a deterministic per-target RNG seed (`42 + rna_idx × 1000`). A target passed Gate B only if R exceeded the 95th percentile of *both* null distributions.

2.5 Gate C: attention-contact correspondence

For each model–target pair, per-head attention matrices were extracted at each layer. Each attention head's matrix was compared to a binary contact matrix derived from ViennaRNA RNAfold 2.7.0 (37°C, dangles 2, Turner 2004 [13]) via Spearman rank correlation. The best-head correlation ρ_S (maximum across all layers and heads) was recorded per model–target pair. A target passed Gate C if $\rho_S \geq 0.10$, a threshold set from noise-floor considerations: for sequences of length 100–400, the standard error of ρ_S under the null is 0.05–0.10.

2.6 Untrained-weight control

To distinguish learned structural representations from architectural inductive bias, the untrained-weight control re-ran Gate C with randomly initialized weights. For each model, pretrained weights were loaded to establish the architecture (layer count, hidden size, intermediate size), then all weight tensors were reinitialized: Xavier uniform for 2D weight matrices, ones for LayerNorm scale, zeros for biases. This preserves positional encodings and attention geometry while removing all learned representations. A five-seed sweep (seeds 0–4, `torch.manual_seed` before reinitialization) quantified initialization variance. The pretrained minus untrained delta on mean ρ_S across targets was computed with a 95% confidence interval using $t(4, 0.025) = 2.776$.

The control was applied to the two DNA models (NTv2 and DNABERT-2) because both showed 16/16 Gate C pass rates, including on negative controls—a pattern suggesting architectural origin. RNA models showed lower and more variable Gate C pass rates, consistent with target-specific learned representations.

2.7 Pre-registered hypotheses

Four hypotheses were specified in the pre-registration document (`preregistration_E1_druggability_gate_v3.md`) and frozen with SHA-256 hash before evaluation:

H-E1 **(Primary)** Gate BA > probing BA with ≥ 0.10 margin, tested via Wilcoxon signed-rank on 6 model-level paired differences (one-sided, $\alpha = 0.05$). DNABERT-2 excluded.

H-E2 Gate-passing models show Spearman $\rho \geq 0.8$ between embedding distance and HTT repeat length (CTL-01/02/03); gate-failing models show $|\rho| < 0.3$.

H-E3 At least one target passes Gate B but fails Gate C (demonstrating the "degeneracy trap"—composition-null survival without structural correspondence).

H-E4 On negative targets, the gate's rejection rate exceeds probing's rejection rate.

2.8 Implementation

All experiments ran on NVIDIA A10G GPUs (24 GB VRAM), one model per container. The full 7-model panel with 1,000 permutations completed in under 60 minutes wall-clock time. Code, pre-registration documents, and all result files are available at [TODO: public repository URL upon submission].

3 Results

3.1 Probing accuracy

All seven models exceed chance on stem-versus-loop classification (Figure 2D; Wilcoxon signed-rank, H_0: median BA = 0.5; $W = 28.0$, $p = 0.008$, $n = 7$). Mean BA across models is 0.686 (range 0.574–0.783). RiNALMo (0.783) and ERNIE-RNA (0.781) are the strongest probes. The two DNA models, NTv2 (0.576) and DNABERT-2 (0.574), are weakest—consistent with cross-domain transfer from DNA pretraining to RNA structure.

3.2 H-E1 (primary): gate discrimination vs. probing

The pre-registration specifies AUC as the discrimination metric. Because the composition-null gate produces a binary PASS/FAIL at the 95th-percentile threshold (a single operating point, not a score), AUC cannot be computed. Balanced accuracy (BA) is the only discrimination metric available at a single threshold and is used here as the realized operationalization.

Gate B BA was computed per model as (sensitivity + specificity) / 2, where sensitivity = fraction of 8 POS targets passing, specificity = fraction of 5 NEG targets rejected.

Model	Sens	Spec	Gate BA	Probing BA	Diff
RNA-FM	0.000	1.000	0.500	0.625	−0.125
RiNALMo	0.125	1.000	0.562	0.783	−0.221
ERNIE-RNA	0.125	1.000	0.562	0.781	−0.219
SpliceBERT	0.250	1.000	0.625	0.743	−0.118
UTR-LM	0.000	1.000	0.500	0.720	−0.220
NTv2	0.125	0.800	0.463	0.576	−0.114

All six differences are negative. Wilcoxon signed-rank (one-sided, H_a: gate > probing): $W = 0.0$, $p = 1.000$, $n = 6$. **H-E1 fails.** The composition-null gate rejects 14–16 of 16 targets per model (Figure 2A), producing sensitivity of 0.000–0.250 and making it a worse discriminator than probing on every model. The gate is too stringent to discriminate druggable from non-druggable targets at $n = 16$.

3.3 Secondary hypotheses

H-E2 predicts that gate-passing models show Spearman $\rho \geq 0.8$ between embedding distance and HTT repeat length (CTL-01/02/03) while gate-failing models show $|\rho| < 0.3$. Gate-passing models satisfy the criterion (4/4, all $\rho = 1.00$), but gate-failing models violate it: both RNA-FM ($\rho = 0.87$) and UTR-LM ($\rho = 1.00$) show strong monotonicity despite failing Gate B. H-E2 fails. Monotonicity is universal—embedding distance tracks repeat length regardless of gate status, a composition effect (longer CAG repeats produce more sequence) rather than a learned structural representation. RNA-FM illustrates this: $\rho = 0.87$ while its actual distances are numerically zero (CTL-02 = 0.000, CTL-03 = 3×10^{-7}), a trivial ordering on noise.

H-E3 predicts at least one target passes Gate B but fails Gate C—mutation sensitivity exceeding the composition null without attention-contact correspondence. Two instances were found, both on SpliceBERT: POS-06 (MALAT1 ENE) and POS-07 (EV71 IRES SLII). H-E3 passes.

These targets survive composition control while the model's attention fails to correspond to the ViennaRNA-predicted contact map—a screening decision based on Gate B alone would accept them.

H-E4 predicts that the gate's rejection rate on negatives exceeds probing's. Gate B rejects 4–5 of 5 negatives per model (80–100%), which trivially exceeds probing's rejection rate under any interpretation. H-E4 is uninformative: Gate B rejects 80–100% of everything, so the directional test is satisfied by blanket stringency rather than selective discrimination.

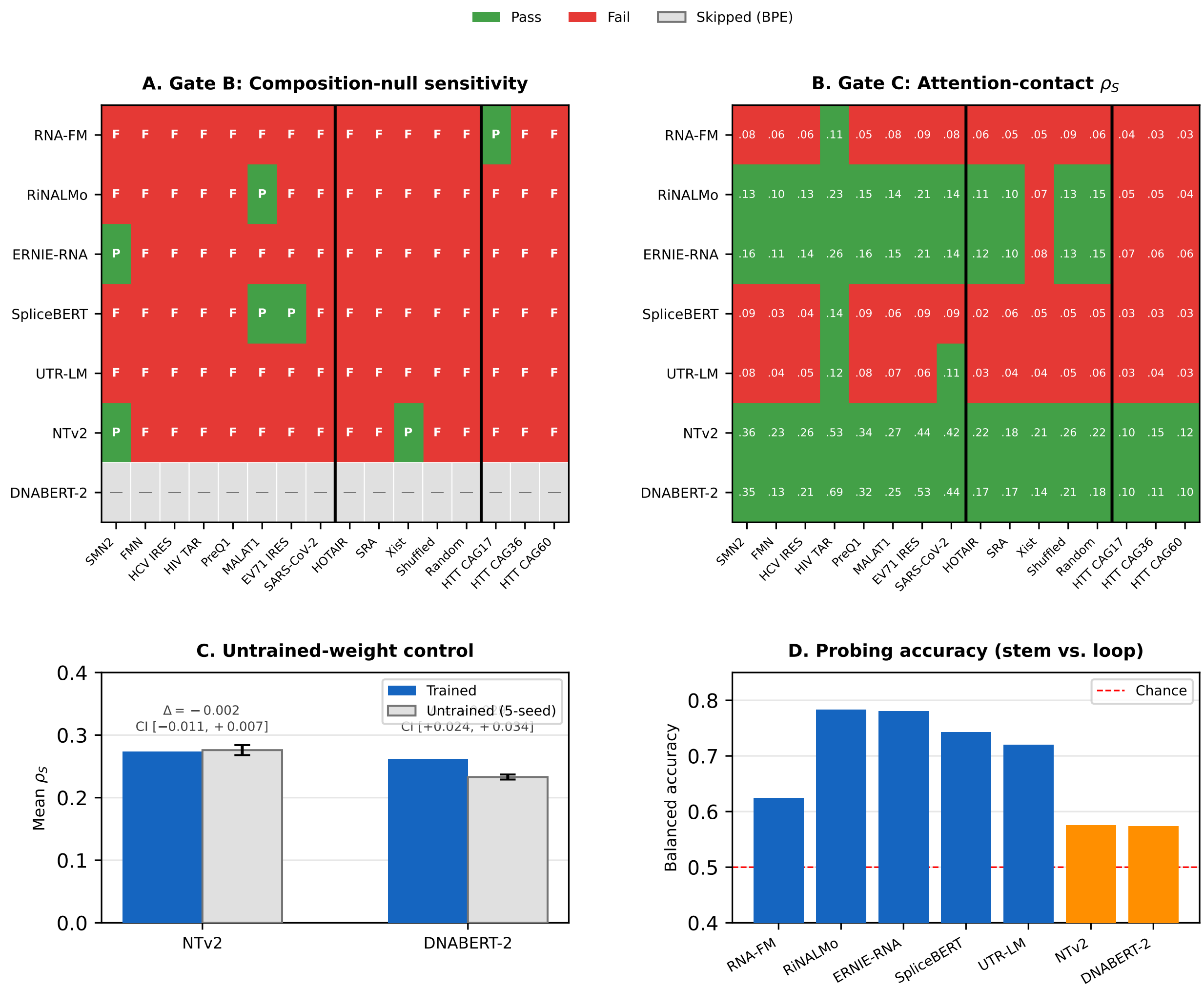

Figure 2: Results overview. **A.** Gate B (composition-null mutation sensitivity): most targets fail across all models; DNABERT-2 is excluded (BPE tokenization). **B.** Gate C (attention–contact ρ_S): NTv2 and DNABERT-2 pass all 16 targets; RNA models show target-specific variation. Values are best-head Spearman ρ_S. **C.** Untrained-weight control: NTv2's attention–contact correlation is fully architectural ($\Delta = -0.002$, CI straddles zero); DNABERT-2 shows a small learned increment ($\Delta = +0.029$). **D.** Probing accuracy (stem vs. loop balanced accuracy): all models exceed chance (0.5); RNA models (blue) outperform DNA models (orange).

3.4 Pre-registered summary

Hypothesis	Result	Interpretation
H-E1 (primary)	FAIL ($p = 1.0$)	Probing outperforms gate on all 6 models
H-E2	FAIL	Monotonicity is universal; gate does not predict it
H-E3	PASS	Degeneracy trap demonstrated (2 instances)
H-E4	UNINFORMATIVE	Gate rejects everything; test trivially satisfied

The primary hypothesis fails. The composition-null gate with 1,000 permutations and 95th-percentile threshold is too conservative to discriminate druggable from non-druggable targets at $n = 16$.

3.5 Architectural artifact in attention-contact correlation

The untrained-weight control exposes a failure mode orthogonal to the pre-registered hypotheses (Figure 2C). NTv2 passes Gate C on all 16 targets (16/16) with trained weights, showing a mean attention-contact correlation of $\rho_S = 0.274$. With randomly initialized weights, it produces $\rho_S = 0.276 \pm 0.008$ (5-seed mean $\pm$ SD) and passes 16/16 on every seed.

Model	Trained	Untrained	Δ	95% CI	Trained C	Untrained C
NTv2	0.274	0.276 ± 0.008	−0.002	$[-0.011, +0.007]$	16/16	16/16
DNABERT-2	0.262	0.233 ± 0.004	+0.029	$[+0.024, +0.034]$	16/16	13–14/16

NTv2's delta is -0.002 with a 95% CI of $[-0.011, +0.007]$, straddling zero. The ESM-style attention architecture produces structure-correlated attention from positional encoding alone. The negative control NEG-04 (a dinucleotide-shuffled tRNA with no real secondary structure) scores $\rho = 0.314$ on untrained NTv2—higher than several positive druggable targets—confirming that the signal is architectural.

DNABERT-2 shows a statistically detectable learned increment (delta = +0.029, 95% CI $[+0.024, +0.034]$, excluding zero across all 5 seeds). The increment is small relative to the untrained baseline: 12% of the base $\rho_S = 0.233$. The untrained model already passes 13–14 of 16 targets; training adds at most 2–3 additional passes. The MosaicBERT architecture produces structure-correlated attention without learned weights; training adds a detectable layer that rarely changes a gate verdict.

Gate C does not discriminate druggable from non-druggable targets in either DNA model. Any evaluation of these models' structural awareness that relies on attention-contact correlation without an untrained-weight control will mistake architectural inductive bias for learned representation.

4 Discussion

The composition-null gate does not discriminate druggable from non-druggable RNA targets. This is a pre-registered negative—the hypothesis, test statistic, and analysis plan were frozen before evaluation—and the negative is as informative as a positive would have been. At 1,000 permutations with a 95th-percentile threshold, most models' per-position mutation sensitivity ratios fall within the range that composition alone predicts. The gate rejects nearly everything, including most positives. Its value is in a different role: a target that survives the composition null carries strong evidence that the model's sensitivity exceeds what nucleotide identity can explain, and the two SpliceBERT targets that pass Gate B but fail Gate C (H-E3) show that composition-immune sensitivity does not guarantee structural correspondence. A screening decision based on mutation sensitivity alone would accept those targets; the two-stage design catches them.

The untrained-weight finding identifies a failure mode orthogonal to composition. NTv2's attention correlates with RNA contact maps at $\rho = 0.274$; random weights produce $\rho = 0.276$ (5-seed CI straddling zero). The ESM-style attention architecture generates structure-correlated patterns from positional encoding alone. NEG-04, a dinucleotide-shuffled tRNA with no real secondary structure, scores $\rho = 0.314$ on untrained NTv2—higher than several positive druggable targets. Any benchmark that uses attention-contact correlation as evidence of structural learning is vulnerable to this confound unless it includes an untrained-weight baseline, a computationally trivial control (one additional forward pass per model). DNABERT-2 shows a statistically real learned increment ($\Delta = +0.029$, CI excludes zero across 5 seeds), but the increment is 12% of the untrained baseline and changes Gate C verdicts on at most 2–3 targets. A continuous metric reporting the pretrained-minus-untrained delta would be more informative than a binary threshold for models in this regime.

These results have direct consequences for how RNA foundation models are evaluated. Per-target gating complements aggregate benchmarks: RiNALMo achieves 78% probing accuracy while passing Gate B on 1 of 16 targets. Aggregate accuracy tells a model developer whether the model has learned structure in general; per-target gating tells a screening team whether the model's confidence on their specific target is credible. Composition control should be standard in mutation sensitivity analyses—the 68.5% stem / 43.9% loop GC asymmetry guarantees that any complement-swap analysis will show a spurious stem-over-loop ratio unless nucleotide identity is stratified out.

Several limitations bound the scope of these conclusions. The 16-target panel constrains statistical power for the primary hypothesis; a larger panel could establish whether a relaxed threshold or modified null improves discrimination. The contact-map ground truth derives from ViennaRNA RNAfold, which may not capture tertiary contacts or context-dependent folds; experimental structure data (SHAPE-MaP, DMS-MaPseq) would provide more reliable ground truth. The untrained-weight control was applied only to the two DNA models; whether architectural artifacts affect RNA-specific architectures at a detectable level remains open. DNABERT-2's BPE tokenization makes its absolute ρ_S values non-comparable to nucleotide-model values—the within-model trained-versus-untrained contrast is valid, but cross-model comparisons involving DNABERT-2 should be interpreted with caution.

The gate functions as artifact-detection infrastructure rather than a druggability predictor—a role that aggregate benchmarks and leaderboard rankings do not fill.

Data availability

Pre-registration documents, all per-model result files (JSON), the 16-target benchmark (`druggability_benchmark_`
and the evaluation code are available at [TODO: public repository URL upon submission] and deposited on Zenodo ([TODO: DOI upon submission]) under a CC-BY-4.0 license. Raw model weights were obtained from HuggingFace Hub under the identifiers listed in Table 2.

Declarations

Ethics approval and consent to participate. This study uses exclusively publicly available pretrained model weights and computationally generated RNA sequences. No individual-level data, human subjects, or animal subjects were involved. No ethical approval was required.

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Authors' contributions. E.T.: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

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